

Kushagra Agarwal

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EDUCATION

Birla Institute of Technology and Science, Pilani
B.E. Chemical Engineering

Aug 2024 – Present
CGPA: 7.04

EXPERIENCE

Research Consultant, WorldQuant

Mar 2025 – Present

- Ranked in the top **2%** of **8,000** consultants (**300K+** applicants), earning a Research Consultant offer.
- Built **66** production alpha models using factor modeling, time-series analysis, and statistical arbitrage for equity markets.
- Reduced alpha-research time by **96%** by automating BRAIN API workflows with Python and Ollama.
- Achieved a **3.54** Sharpe ratio with **21.42%** returns and **2%** drawdown through a proprietary alpha for China's equity market.

Intern, MD Enterprises Global

May 2026 – Jul 2026

- Established the company's first institutional fundraising pipeline and led a team targeting INR **10 lakh** in funding.
- Wrote zero-cost PPP rollout go-to-market reports for securing a contract with the Ministry of Road Transport and Highways.
- Developed ML-based face and voice authentication for **1.1 million** NSDL Jiffy users, replacing Aadhaar OTP verification.

PROJECTS

STRATOS (Actor-Based Flight Software)

[GitHub](#)

- Architected an actor-based flight-software framework that decouples control logic from real-time sensor pipelines.
- Designed an event-driven message-passing system supporting **1B+** states with deterministic execution across flight modes.
- Implemented inheritance-based subscription model enabling **10+** reader modules to dynamically subscribe to sensor streams.

Transformer Language Model Ablation Lab

[GitHub](#)

- Built a decoder-only Transformer LM (**6L, 512d, 8 heads**) with a controlled single-axis ablation framework on WikiText-2.
- Engineered a registry to swap attention, positional-encoding, and architecture variants, isolating each design change.
- Benchmarked BPE-level perplexity, loss, throughput, and memory via config-hashed, append-only experiment records.

Single-Image 3D Reconstruction

[GitHub](#)

- Built a pipeline that reconstructs 3D meshes from a single RGB image using GLPN monocular depth estimation.
- Converted depth maps to RGBD point clouds via Open3D camera projection, with outlier removal and normal estimation.
- Generated surface meshes through Poisson reconstruction and deployed the pipeline behind an interactive web app.

POSITIONS OF RESPONSIBILITY

President, Kalipatnapu AR-VR Lab, BITS Pilani

May 2026 – Present

- Founded the lab's AI/ML and XR research vertical, defining its technical roadmap and scaling the team from the ground up.
- Led computer-vision and 3D Gaussian Splatting research, directing project scope and mentoring associates.

Teaching Assistant, International Programmes & Collaboration Division

Jan 2026 – Present

- Organized a DAAD information session with Indian diplomats from Germany.
- Managed applications and documentation for international scholarships for **five** students pursuing overseas programmes.

Founder & President, Board Games Club

Jul 2025 – Present

- Organized a two-day, three-stage tournament with **50+** registrations and INR **55K+** in cash sponsorship.
- Hosted **10+** open sessions reaching **500+** participants and managed INR **30K+** in inventory.

ACTIVITIES & VOLUNTEERING

Pilani Innovation and Entrepreneurship Development Society Student Team

- Secured **50+** partnerships and INR **30 lakh** in cash sponsorships for Ignite '25, a six-day accelerator programme.
- Led **30** volunteers to organize Solve for Pilani, generating INR **70K** for five local businesses and attracting **3,000+** visitors.
- Built a full-stack event platform used by **600+** students, sustaining approximately **1,000** daily visits through gamification.

Debating Society, BITS Pilani

- Best Speaker and fourth-best team at Gambit, India's largest APD Freshers' Tournament, hosted by SRCC.
- Novice Grand Finalist and fourth-best team at IIT Roorkee's annual BPD Tournament.

RISE — Tutor & Research Mentor

- Improved a student's TMUA score from **2/9** to **7/9**, supporting their admission to Imperial College London.
- Designed **10+** full-length TMUA mock exams with typeset LaTeX material and targeted feedback.
- Mentored a student across **30+** research sessions to a publication in the International Journal for Multidisciplinary Research.